

Topology-Driven Edge Predictions with Graph Machine Learning for Optical Network Growth

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Abstract: Graph representation learning on real-world optical core networks outperforms edge prediction heuristics by 10 times, achieving up to 93.4% accuracy on BT(UK), COST(EU), and CORONET(USA) by learning from 10% training data. © 2023 The Author(s)

1. Introduction: A Graph-Based Representation of Communication Systems

Optical network topologies can be represented as graphs, with networking equipment facilities as nodes and fiber cables as edges [1]. Edge or link prediction is the process of adding new fiber cables on these topologies to enable proactive capacity planning and automated, optimised expansion of optical networks [2, 3, 4]. In industrial settings, extended over-utilization of a link over months often signals the need for physical fiber expansion. However, the physical layout and interconnections in the network topology graph also provide meaningful signals about growth in the absence of sensitive real-time data. Our model aims to learn from the stable, informative structural patterns within mature, well-performing optical networks, which have implicit historical growth and expansion data that are valuable for predicting future expansions without directly considering traffic volume. Further, the distinct characteristics of optical networks, compared to social networks, pose unique challenges for standard link prediction heuristics, which make localized assumptions, whereas our model aims to overcome these limitations by learning global patterns. We discover and find the following: (i) Through strategic utilisation of the structural and spatial patterns of optical networks, we predict future inter-nodal fiber connections, as evidenced by a 98.5% accuracy on COST with 80% training data and 93.4% on BT with 10% training data. Remarkably, all topologies can learn up to 88-93% of network design from merely 10% of training data. (ii) Graph-based embeddings passed onto machine learning models demonstrably advance network design intelligence, outperforming statistical edge prediction techniques by multiple orders of magnitude as seen for COST (10 times), BT (20 times), and CORONET (20 times). (iii) The methodology empowers and enables network operators with efficient automated decision-making tools, as the learning and inference process takes less than a minute. To the best of our knowledge, we are the first to introduce graph-based machine learning (ML) for edge prediction in optical networks.

2. Proposed Method: Graph Embeddings Passed onto Machine Learning Classifier for Edge Prediction

Link Prediction: Given a network graph $G = (V, E)$ where V is the set of nodes and E is the set of edges, the goal is to estimate a scoring function $f : V \times V \rightarrow \mathbb{R}$ that assigns a score to potential edges (x, y) , not in E . The score $f(x, y) \rightarrow \mathbb{R}$, where $(x, y) \notin E$ indicates the likelihood of a link forming between nodes u and v , with higher scores implying a higher probability of linkage. Function f can be based on different mathematical models and algorithms that encode topological and nodal attributes of the graph G to predict non-observed links. Predicted scores are used to rank potential links and evaluate against ground truth links.

Node2Vec: Node2vec [3] is a node embedding technique designed to transform complex network structures into meaningful and compact vector representations using biased random walks. (i) **Random Walks:** Starting from each node in the graph, node2vec initiates a series of random paths or walks. Each step in the path is influenced by a balance between exploration and exploitation, driven by parameters p and q . When $p > q$, the walks are inclined towards breadth-first exploration. When $q > p$, the walks lean towards depth-first exploration. If $p = q$, the process is similar to a standard random walk. (ii) **Transition Probabilities:** The likelihood of moving from one node to another is based on edge weights and neighboring nodes. The likelihood ensures that the walk is not purely random but is influenced by the structure of the graph. (iii) **Embedding Generation:** Using the Skip-gram model from the Word2Vec architecture [3], the paths generated from random walks are seen as

‘sentences’ and the nodes as ‘words’. The aim is to predict neighboring nodes in a walk which results in nodes having d -dimensional vector representations that encapsulate their neighbourhood within the graph.

Predictive Modelling Logistic Regression: Node2vec embeddings are passed onto a binary classifier to predict the existence of an edge between nodes using supervised machine learning on graphs. (i) **Model:** The probability of an edge between nodes is given by $p = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$, where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the logistic function where x_1 is the dot product of node embeddings, x_2 represents link lengths in km and $\theta_0, \theta_1, \theta_2$ are the model parameters (bias and coefficients). (ii) **Training:** We minimize the logistic loss, $L(y, p) = -[y \log(p) + (1-y) \log(1-p)]$, where y is the true label and p is the predicted probability. (iii) **Evaluation:** Model is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) and accuracy.

Comparison with Local, Global and Random Heuristics: *Common Neighbors (CN)* measures the number of neighbours two nodes share using $f_{CN}(x, y) = |\Gamma(x) \cap \Gamma(y)|$ where $\Gamma(x)$ denotes the set of neighbours of node x [5]. *Jaccard Coefficient (JC)* measures the proportion of shared neighbours using $f_{JC}(x, y) = \frac{|\Gamma(x) \cap \Gamma(y)|}{|\Gamma(x) \cup \Gamma(y)|}$ [5]. *Adamic-Adar Index (AA)* gives more weight to shared neighbors that have fewer links, $f_{AA}(x, y) = \sum_{z \in \Gamma(x) \cap \Gamma(y)} \frac{1}{\log |\Gamma(z)|}$ [5]. *Rooted PageRank (RPR)* extends PageRank by calculating a stationary distribution, π_x , for a random walk originating from a node x , moving to adjacent nodes with probability α , or reverting to x with probability $1 - \alpha$. The distribution, specifically $[\pi_x]_y$ at node y , aids in predicting the link (x, y) . In undirected networks, a symmetrical RPR variant employs $f_{RPR}(x, y) = [\pi_x]_y + [\pi_y]_x$ for link prediction [5]. *Random* serves as a control baseline where links are predicted by sampling from the list of non-edges [2].

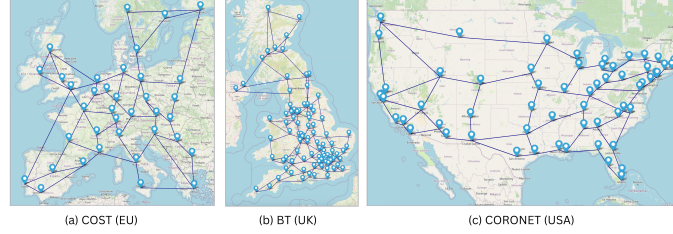


Fig. 1. Core Optical Networks as Graphs (a) COST (b) BT (c) CORONET

Dataset and Implementation Details: We examine the growth of optical networks, focusing on BT (UK, $|V|=106$, $|E|=180$), COST (Europe, $|V|=37$, $|E|=57$), and CORONET (USA, $|V|=60$, $|E|=75$), characterized by locations, topological structure, and link lengths (Fig 1) [1, 2, 6]. Each topology is represented as a graph and modelled as a separate machine-learning pipeline (Fig 2(b)). The positive edges represent existing edges in the network, and an equal number of negative edges are sampled from the set of non-edges in the network for the creation of a balanced dataset. The model learned only from a specific portion of the graph as it was tested on different test sizes from 0.1 to 0.9 at intervals of 0.1. Node embeddings were learned using Node2Vec (dimension = 64, walks = 200, length = 30, p=1, q=1) and passed onto a Logistic Regression binary classifier (solver=lbfgs, regularisation= L2, C=1, maximum iterations=100). Similarly, a portion of edges were removed from a network from 0.1 to 0.9, and the heuristics were computed and compared against the model on the metric accuracy.

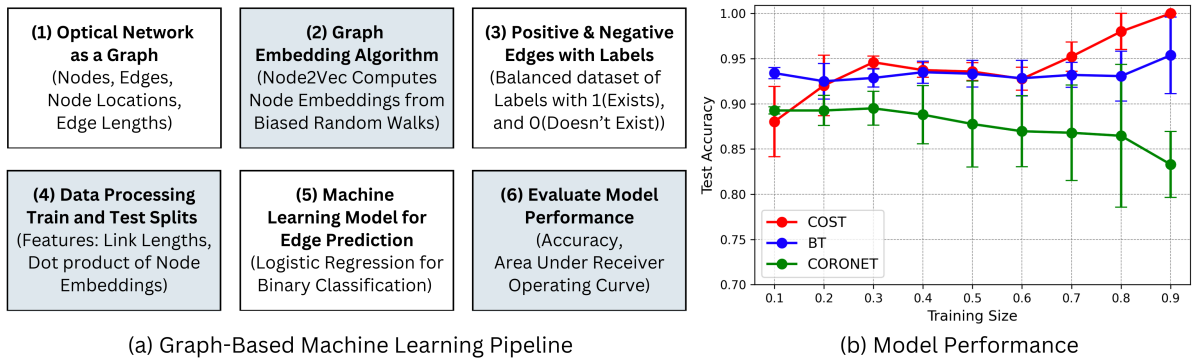


Fig. 2. Model Pipeline and Evaluation of Edge Prediction on Optical Networks

3. Results: Evaluation, Interpretation, and Benchmarking of Link Prediction Methods

We quantify the *strong performance* of models that learn the structural patterns of an optical network to predict potential future inter-nodal connections. COST topology has the best performance averaged across 3 random seeds with respect to the AUC-ROC and accuracy (0.99, 98.4%) followed by BT(0.98, 93.1%) and CORONET(0.92, 86.5%) when learned from 80% training data. The model is *scalable*, works with diverse graph sizes of 37, 60 and 106 and is *adaptable* to both the US and the EU. We also find the *robustness* of the model at different training sizes predicting 93.4% at only 10% of training data for BT. The pattern follows across three topologies, the model is able to predict up to 88.1-93.4% even on a 10% training dataset. The small accuracy gain for BT and COST from 10% to 90% training data is restrained by the small graph size, as these methods are optimized for larger graphs with thousands of nodes and a hundred thousand edges (Fig 2(a)). Logistic regression struggles with non-linear data relationships, necessitating more complex models in the case of CORONET. Our model *significantly outperforms* standard link prediction techniques by multiple orders of magnitude; for instance, our model predicts approximately 10 times better at 88% than heuristics at 7.8% on COST topology from merely 10% of network data (Table 1). The model appears to learn latent features of the topology not captured by heuristics, as evidenced by the significant performance gap. Heuristics make assumptions about network structure and growth, such as reliance on local neighbourhood information and growth being random and stochastic. Our model learns from the organized, planned and strategic expansion of optical infrastructure networks based on traffic demands, geographic factors, and budget constraints through its topological structure. The complete machine-learning pipeline takes less than 1 minute and is *efficient* enough to empower network operators.

Table 1. Model Comparison with Heuristics (Maximum Value is compared) at training set t

Topology	Baseline (t = 0.5)	Accuracy (t = 0.5)	Baseline (t = 0.1)	Accuracy (t = 0.1)
COST	0.0714 (Pagerank)	0.94 ± 0.01	0.0784 (Adamic Adar)	0.88 ± 0.04
BT	0.1222 (Adamic Adar)	0.93 ± 0.01	0.0432 (Jaccard)	0.93 ± 0.01
CORONET	0.0256 (Jaccard)	0.88 ± 0.05	0.0423 (Random)	0.89 ± 0.00

4. Conclusion: Graph-based Machine Learning Can Efficiently Automate Network Planning and Growth

Topological-data-driven and graph-based machine learning approaches for network planning on three diverse optical network topologies outperform conventional edge prediction methods by 10-20 times. Our model has learned from topologies which have historically grown in size from monitoring traffic and forecasting demand. It attains test accuracy of 98.5% on COST with 80% training data, and 93.4% on BT with a mere 10% of training data, thereby exhibiting scalability, robustness, adaptability and efficiency. Our study makes automated network design at the physical layer feasible both computationally and economically, further enhancing engineer productivity through modelling spatial and structural attributes to recommend optical fiber additions as edges to the network.

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References

- [1] S. Orlowski, R. Wessälly, M. Pióro, A. Tomaszewski, *Sndlib 1.0—Survivable Network Design Library*, Networks **55**(3), 276–286 (2010).
- [2] Y.-Z. Xu, D. Saad, *Network Pruning and Growth: Probabilistic Optimization*, Physical Review Research **5**(3), 033087 (2023).
- [3] A. Grover, J. Leskovec, *node2vec: Scalable Feature Learning for Networks*, in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, (2016), pp. 855–864.
- [4] R. Matzner, D. Semrau, R. Luo, G. Zervas, P. Bayvel, *Making Intelligent Topology Design Choices: Understanding Structural & Physical Property Performance Implications in Optical Networks*, Journal of Optical Communications and Networking **13**(8), D53–D67 (2021).
- [5] M. Zhang, *Graph Neural Networks: Link Prediction*, in *Graph Neural Networks: Foundations, Frontiers, and Applications*, L. Wu, P. Cui, J. Pei, L. Zhao, eds. (Springer Singapore, 2022), pp. 195–223.
- [6] P. Wright, R. Davey, A. Lord, *Cost Model Comparison of ZR/ZR+ Modules Against Traditional WDM Transponders for 400G IP/WDM Core Networks*, in *2020 European Conference on Optical Communications (ECOC)*, (IEEE, 2020), pp. 1–4.